

Carl Chua

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EXPERIENCE

Software Engineer

2025 - Current

Labelbox

San Francisco, CA

- Developed a full-stack RL evaluation platform for agent evaluation and benchmarking. Built a configurable grading engine and migrated execution from Modal to GKE. Implemented a sidecar proxy to intercept outbound agent traffic, securely injecting API credentials for protected, isolated evaluation and controlled API access.
- Improved platform reliability via submission concurrency limiting, stuck-run detection, and token-bucket rate limiting. Improved developer productivity with a git worktree picker.
- Worked with FDEs to design custom RL environments for model evaluation and finetuning.
- Built a cross-platform video collection app and GCP processing pipeline for robotics RL training, supporting 10K+ streamed chunk uploads/day. Ensured IMU data to video alignment across chunked recordings were accurate to within 50ms and achieved 99% eventual upload reliability via retry and recovery mechanisms. Added e2e observability and integrated Firebase Cloud Messaging as a notification system.

AI Engineer

2022 - 2025

BlackRock

San Francisco, CA

- Built an agentic framework, setting up gRPC-based plugin servers to integrate multiple agents into a single Copilot.
- Implemented RAG-based plugins. Processed and cleaned data to enhance synthetic data generation for fine-tuning embedding models, speeding up training by 15%
- Wrote evaluation suites to ensure 96%+ orchestration accuracy when onboarding new agents. Built out LLM-as-a-judge custom evaluators to ensure specific Copilot behaviors.
- Improved observability by adding support to log Langchain execution traces to Grafana, while filtering sensitive PII data from logs to maintain security and compliance.
- Implemented streaming functionality on a Java-based server to improve user access to financial data

Machine Learning Intern

2021

Rimble Esports Analytics

Berkeley, CA

- Created custom ML models to predict realtime and pregame statistics for e-sports. Did data cleaning to create training/test data. Outperformed existing betting numbers.

Computer Science Researcher

2020-2021

Swarm Lab, UC Berkeley

Berkeley, CA

- Used OpenCV, homography matrices, and feature detection techniques such as SIFT to help detect and control state changes by re-orienting images at off-angles. Implemented an ML pipeline, using JSON files to automate image extraction from videos and using CNNs for image classification. Worked on applying transfer learning to the current CNN model to improve performance.

EDUCATION

University of California, Berkeley

GPA: 3.75

B.S. Electrical Engineering & Computer Science (EECS)

Data Structures, Computer Architecture, Artificial Intelligence, Advanced Algorithms & Intractable Problems, Probability & Random Processes, Operating Systems, Machine Learning, Robotics, Deep Reinforcement Learning*

*Graduate-level course

TECHNICAL SKILLS

Languages: Python, Java, C, Golang, C++, Matlab, SQL, HTML/CSS

Technologies: gRPC, GCP, DataDog, NumPy, Linux, Keras, TensorFlow, PyTorch, Selenium, Pandas, Git, Docker, Kubernetes, Grafana